Exactness as a Lens: Chain Complexes, Mayer–Vietoris, and Long Exact Sequences

A Computation-First Homological Atlas

June 8, 2026

A computation-first guide to understanding a mathematical situation by cutting, comparing, connecting, and gluing.

A chain complex records what fails to be exact. A short exact sequence compares three worlds. A long exact sequence is the shadow that comparison casts on cohomology. Mayer–Vietoris is the art of understanding a global object by cutting it into local pieces and measuring the obstruction to gluing.

1 The Situation and the Targets

The situation. We study

$$X = \mathbb{P}^1_{\mathbb{C}},$$

which we read in *two* compatible ways. Topologically $\mathbb{P}^1_{\mathbb{C}}$ is the real 2-sphere S^2, so it carries de Rham cohomology over $\mathbb{R}$. Analytically it is a complex curve carrying the sheaves $\mathcal{O}$ (holomorphic functions), $\mathcal{O}^\times$ (nowhere-zero holomorphic functions), the constant sheaf $\mathbb{Z}$, and the line bundles $\mathcal{O}(d)$. Both readings use the *same* two-set cover

$$U_0 = \mathbb{P}^1 \setminus \{\infty\} \cong \mathbb{C}, \qquad U_\infty = \mathbb{P}^1 \setminus \{0\} \cong \mathbb{C}, \qquad U_0 \cap U_\infty = \mathbb{C}^\times \simeq S^1,$$

with z the coordinate on U_0 and $w = 1/z$ the coordinate on U_∞. The pieces U_0, U_∞ are contractible (and Stein); the overlap $\mathbb{C}^\times$ is homotopy equivalent to a circle. This single picture — two disks glued along an annulus — is the engine of the whole document. Figure 1 shows this cover both as a sphere and as a Venn diagram.

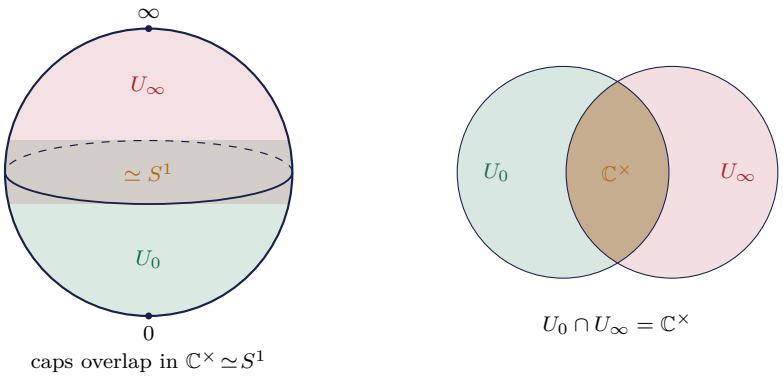


Figure 1. *The cover of $\mathbb{P}^1 = S^2$: two contractible caps U_0 (around 0) and U_∞ (around ∞) overlapping in the equatorial annulus $\mathbb{C}^\times \simeq S^1$ — the homotopy model that drives every computation below.* Left*: as a sphere;* right*: as a Venn diagram. (Supports §1, §7, §8.)*

Remark 1 (Stated assumptions)**.** The situation slot was delivered empty. We fix $X = \mathbb{P}^1_{\mathbb{C}}$ because it lets *every* homological tool below be computed in closed form and unifies all four requested micro-examples (S^1, S^2, $\mathbb{P}^1$, line bundles) in one space. De Rham statements use $\mathbb{R}$ coefficients; Čech/sheaf statements use the indicated sheaf. We use the cover $\{U_0, U_\infty\}$ throughout; it is a *Leray* (acyclic) cover for the coherent sheaves below, so Čech cohomology equals sheaf cohomology. Where a surjectivity uses a smooth partition of unity we say so explicitly, and we flag where the holomorphic world has none.

The targets of understanding.

1. the relevant chain/cochain complexes (de Rham on X; Čech on $\{U_0, U_\infty\}$);
2. the local pieces U_0, U_∞, and the overlap $U_0 \cap U_\infty = \mathbb{C}^\times$;
3. the short exact sequence comparing global data with local data;
4. the Mayer–Vietoris sequence;
5. the connecting homomorphism δ;
6. the resulting long exact sequence;
7. the concrete groups $H^\bullet_{\mathrm{dR}}(\mathbb{P}^1)$, $H^\bullet(\mathbb{P}^1, \mathcal{O})$, $\mathrm{Pic}(\mathbb{P}^1)$;
8. the geometric meaning of every kernel, image, cokernel, and obstruction.

We do not treat homological algebra as formal decoration. We use it as a microscope: each exact sequence is pointed at a specific target and made to reveal it.

2 Chain Complexes: The Basic Machine

The de Rham cochain complex. On the smooth 2-manifold $X = \mathbb{P}^1 \cong S^2$,

$$0 \longrightarrow \Omega^0(X) \xrightarrow{d} \Omega^1(X) \xrightarrow{d} \Omega^2(X) \longrightarrow 0,$$

where $\Omega^k(X)$ is the $\mathbb{R}$-vector space of smooth k-forms and d is the exterior derivative. In a local coordinate, Ω^0 is functions f, Ω^1 is $a\,dx + b\,dy$, Ω^2 is $h\,dx \wedge dy$, and

$$d^2 = 0 \qquad (\text{e.g. } d(df) = \partial_{xy} f\, dx \wedge dy - \partial_{yx} f\, dy \wedge dx = 0).$$

A form ω is a *cocycle* (closed) if $d\omega = 0$, a *coboundary* (exact) if $\omega = d\eta$. The cohomology is

$$H^n_{\mathrm{dR}}(X) = \frac{\ker(d : \Omega^n \to \Omega^{n+1})}{\mathrm{im}(d : \Omega^{n-1} \to \Omega^n)} = \frac{\text{closed } n\text{-forms}}{\text{exact } n\text{-forms}}.$$

The Čech cochain complex. For the two-set cover $\mathfrak{U} = \{U_0, U_\infty\}$ and a sheaf $\mathcal{F}$,

$$0 \longrightarrow \check{C}^0(\mathfrak{U}, \mathcal{F}) \xrightarrow{\delta} \check{C}^1(\mathfrak{U}, \mathcal{F}) \longrightarrow 0, \qquad \check{C}^0 = \mathcal{F}(U_0) \oplus \mathcal{F}(U_\infty), \quad \check{C}^1 = \mathcal{F}(U_0 \cap U_\infty),$$

with

$$\delta(f_0, f_\infty) = (f_\infty - f_0)\big|_{U_0 \cap U_\infty}.$$

Because the cover has only two sets there are no triple overlaps, so $\check{C}^{\geq 2} = 0$ and $\delta^2 = 0$ *trivially*; the cocycle condition in degree ≥ 2 is vacuous. Then

$$\check{H}^0(\mathfrak{U}, \mathcal{F}) = \ker \delta = \{(f_0, f_\infty) : f_0 = f_\infty \text{ on } U_0 \cap U_\infty\} = \mathcal{F}(X), \qquad \check{H}^1(\mathfrak{U}, \mathcal{F}) = \mathrm{coker}\, \delta.$$

Cohomology measures closed objects modulo exact objects.

Concretely on $\mathbb{C}^\times$: $d\theta$ is closed but not exact (θ is not a global function), so $[d\theta] \neq 0$ in $H^1_{\mathrm{dR}}(\mathbb{C}^\times)$. That one nonzero class is the seed from which the H^2 of $\mathbb{P}^1$ will grow. The overlap and its winding class are drawn in Figure 2.

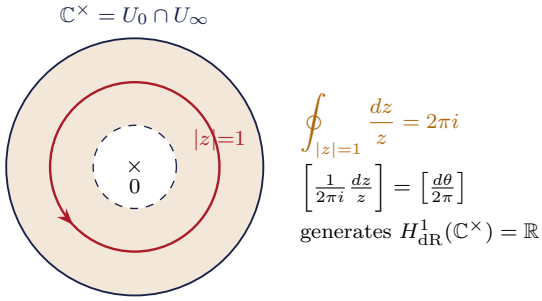


Figure 2. *The overlap $\mathbb{C}^\times \simeq S^1$ as a punctured disk. The oriented loop $|z| = 1$ carries the winding class $[\frac{1}{2\pi i}\frac{dz}{z}] = [\frac{d\theta}{2\pi}]$, the single generator of $H^1_{\mathrm{dR}}(\mathbb{C}^\times) = \mathbb{R}$ that the connecting map will promote to the area class of $\mathbb{P}^1$. (Supports §2, Example C.)*

3 Exactness as a Diagnostic

A pair of maps is *exact* at the middle term when image equals kernel:

$$A \xrightarrow{f} B \xrightarrow{g} C \quad \text{is exact at } B \quad \Longleftrightarrow \quad \operatorname{im} f = \ker g.$$

We read each exact sequence as a measurement:

- $\ker g$ collects the elements of B that satisfy the *compatibility constraint* imposed by g;
- $\operatorname{im} f$ collects the elements of B that are *already produced* by A ("already glued");
- exactness says: *every* constrained element is produced — no more, no less;
- failure of exactness, measured by $\ker g / \operatorname{im} f$, is an *obstruction*: a constrained element that is *not* produced.

Every exact sequence in this document is accompanied by its maps, its kernel, its image, its connecting map (when present), and a one-line diagnosis of what the obstruction means.

4 Short Exact Sequences of Complexes

A *short exact sequence of cochain complexes* is a triple of complexes with degreewise maps

$$0 \longrightarrow A^\bullet \xrightarrow{i} B^\bullet \xrightarrow{p} C^\bullet \longrightarrow 0$$

such that in every degree the row is exact and the maps commute with the differentials:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A^n & \xrightarrow{i^n} & B^n & \xrightarrow{p^n} & C^n & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle d_A} & & \downarrow{\scriptstyle d_B} & & \downarrow{\scriptstyle d_C} & & \\ 0 & \longrightarrow & A^{n+1} & \xrightarrow{i^{n+1}} & B^{n+1} & \xrightarrow{p^{n+1}} & C^{n+1} & \longrightarrow & 0 \end{array}$$

Exactness of each row says: i^n is injective, $\operatorname{im} i^n = \ker p^n$, and p^n is surjective. Commutativity $d_B i = i\, d_A$ and $p\, d_B = d_C\, p$ says the three differentials are *one differential seen in three worlds.* This is exactly what lets the cohomologies communicate.

Diagnosis. *$A^\bullet$ is the sub-world sitting inside $B^\bullet$; $C^\bullet$ is the quotient world $B^\bullet/A^\bullet$. The sequence compares "global" and "local" data in our case: $A^\bullet = \Omega^\bullet(X)$, $B^\bullet = \Omega^\bullet(U_0) \oplus \Omega^\bullet(U_\infty)$, $C^\bullet = \Omega^\bullet(\mathbb{C}^\times)$ (Section 7).*

Proposition 1 (Zig-zag)**.** *A short exact sequence of complexes induces a long exact sequence in cohomology,*

$$\cdots \to H^n(A^\bullet) \xrightarrow{i_*} H^n(B^\bullet) \xrightarrow{p_*} H^n(C^\bullet) \xrightarrow{\delta} H^{n+1}(A^\bullet) \to \cdots,$$

where δ is the connecting homomorphism constructed in Section 5.

5 The Connecting Homomorphism

We *construct* $\delta\colon H^n(C^\bullet) \to H^{n+1}(A^\bullet)$; we do not merely name it. The construction is a chase through the diagram

$$\begin{array}{ccccc} & & B^n & \xrightarrow{p^n} & C^n \\ & & \downarrow{\scriptstyle d_B} & & \downarrow{\scriptstyle d_C} \\ A^{n+1} & \xrightarrow[i^{n+1}]{} & B^{n+1} & \xrightarrow{p^{n+1}} & C^{n+1}. \end{array}$$

1. Start with a class $[c] \in H^n(C^\bullet)$, represented by a cocycle $c \in C^n$ with $d_C c = 0$.
2. Since p^n is surjective, choose a lift $b \in B^n$ with $p^n(b) = c$.
3. Push down: $d_B b \in B^{n+1}$.
4. Compute its image: $p^{n+1}(d_B b) = d_C\, p^n(b) = d_C c = 0$, using commutativity and step 1.
5. Hence $d_B b \in \ker p^{n+1} = \operatorname{im} i^{n+1}$ by exactness.
6. Since i^{n+1} is injective, there is a *unique* $a \in A^{n+1}$ with $i^{n+1}(a) = d_B b$.
7. Define $\delta([c]) = [a] \in H^{n+1}(A^\bullet)$.

a is a cocycle. Apply i^{n+2} to $d_A a$: by commutativity $i^{n+2}(d_A a) = d_B\, i^{n+1}(a) = d_B(d_B b) = 0$. As i^{n+2} is injective, $d_A a = 0$.
Independence of the lift b. Another lift is $b' = b + i(a')$ for some $a' \in A^n$ (since $p(b' - b) = 0$). Then $d_B b' = d_B b + i(d_A a')$, so the new a is $a + d_A a'$, and $[a + d_A a'] = [a]$.
Independence of the representative c. Replacing c by $c + d_C\gamma$ ($\gamma \in C^{n-1}$): lift γ to $\beta \in B^{n-1}$, then $b + d_B\beta$ lifts $c + d_C\gamma$ and $d_B(b + d_B\beta) = d_B b$, so a is unchanged. Thus δ is a well-defined homomorphism.

> *The connecting homomorphism measures the obstruction to lifting a closed object in $C^\bullet$ to a closed object in $B^\bullet$.*

Diagnosis. *In our Mayer–Vietoris setting, $C^\bullet = \Omega^\bullet(\mathbb{C}^\times)$: δ takes a closed form on the overlap $\mathbb{C}^\times$ and returns the global class on $\mathbb{P}^1$ obstructing the splitting of that overlap form into a* difference of *forms living on U_0 and U_∞. A nonzero δ means the overlap form cannot be untwisted into local pieces.*

6 The Long Exact Sequence

Proposition 1 gives, for any short exact sequence of complexes,

$$\cdots \to H^n(A^\bullet) \xrightarrow{i_*} H^n(B^\bullet) \xrightarrow{p_*} H^n(C^\bullet) \xrightarrow{\delta} H^{n+1}(A^\bullet) \xrightarrow{i_*} H^{n+1}(B^\bullet) \to \cdots$$

Read segment by segment:

- *Exact at $H^n(B)$:* $\operatorname{im} i_* = \ker p_*$. A class on B dies in C iff it comes from A.
- *Exact at $H^n(C)$:* $\operatorname{im} p_* = \ker \delta$. A class on C lifts to B iff its obstruction δ vanishes.
- *Exact at $H^{n+1}(A)$:* $\operatorname{im} \delta = \ker i_*$. A class on A is a genuine obstruction (in the image of δ) iff it becomes trivial in B.

We now specialize the abstract A, B, C to the geometric pieces of $\mathbb{P}^1$.

7 Mayer–Vietoris from a Short Exact Sequence

Let $X = U \cup V$; here $U = U_0$, $V = U_\infty$, $U \cap V = \mathbb{C}^\times$. The *Mayer–Vietoris short exact sequence of de Rham complexes* is

$$0 \longrightarrow \Omega^\bullet(X) \xrightarrow{r} \Omega^\bullet(U) \oplus \Omega^\bullet(V) \xrightarrow{s} \Omega^\bullet(U \cap V) \longrightarrow 0,$$

$$r(\omega) = (\omega|_U,\ \omega|_V), \qquad s(\alpha, \beta) = \alpha|_{U\cap V} - \beta|_{U\cap V}.$$

Exactness, checked term by term.

- r *injective.* If $\omega|_U = 0$ and $\omega|_V = 0$ then $\omega = 0$, since $\{U, V\}$ covers X and a form is determined by its restrictions. ($\ker r = 0$: *compatibility forces nothing extra.*)
- $s \circ r = 0$. $s(r\omega) = \omega|_{U\cap V} - \omega|_{U\cap V} = 0$: the two restrictions of a global form agree on the overlap.
- $\ker s = \operatorname{im} r$. If $s(\alpha, \beta) = 0$ then $\alpha = \beta$ on $U \cap V$, so α, β *glue* to a global form ω with $r\omega = (\alpha, \beta)$. (This is the gluing axiom in algebraic dress.)
- s *surjective.* Given $\gamma \in \Omega^k(U \cap V)$, pick a smooth partition of unity $\{\rho_U, \rho_V\}$ subordinate to $\{U, V\}$ ($\rho_U + \rho_V = 1$, $\operatorname{Supp} \rho_U \subset U$, $\operatorname{Supp} \rho_V \subset V$). Set

  $$\alpha = \rho_V\, \gamma \text{ (extended by 0 to } U), \qquad \beta = -\rho_U\, \gamma \text{ (extended by 0 to } V).$$

  These are smooth because ρ_V vanishes near $U \setminus V$ and ρ_U vanishes near $V \setminus U$. On the overlap $\alpha - \beta = (\rho_U + \rho_V)\gamma = \gamma$, so $s(\alpha, \beta) = \gamma$.

Remark 2 (Where the partition of unity is essential)**.** Surjectivity of s is the *only* step needing a partition of unity, and it is genuinely smooth: there is **no holomorphic partition of unity**. For holomorphic data one must instead control surjectivity by hand or pass to the Čech complex, where δ onto $\check{C}^1$ is automatic because $\check{C}^1$ is the top term (Section 2). This is the precise point where the smooth and holomorphic stories diverge.

Feeding this short exact sequence into Proposition 1 gives the **Mayer–Vietoris long exact sequence**

$$\cdots \to H^k_{\mathrm{dR}}(X) \xrightarrow{r_*} H^k_{\mathrm{dR}}(U) \oplus H^k_{\mathrm{dR}}(V) \xrightarrow{s_*} H^k_{\mathrm{dR}}(U \cap V) \xrightarrow{\delta} H^{k+1}_{\mathrm{dR}}(X) \to \cdots$$

For sheaf cohomology the analogous sequence (no partition of unity) is the Čech–Mayer–Vietoris sequence

$$0 \to \mathcal{F}(X) \to \mathcal{F}(U) \oplus \mathcal{F}(V) \to \mathcal{F}(U \cap V) \to H^1(X, \mathcal{F}) \to H^1(U, \mathcal{F}) \oplus H^1(V, \mathcal{F}) \to \cdots$$

whose maps are $f \mapsto (f|_U, f|_V)$ and $(f, g) \mapsto f|_{U\cap V} - g|_{U\cap V}$.

8 Explicit Mayer–Vietoris Computation for $\mathbb{P}^1$

Cohomology of the pieces. $U_0 \cong \mathbb{C}$ and $U_\infty \cong \mathbb{C}$ are contractible, so

$$H^0_{\mathrm{dR}}(U_0) = H^0_{\mathrm{dR}}(U_\infty) = \mathbb{R}, \qquad H^{\geq 1}_{\mathrm{dR}}(U_0) = H^{\geq 1}_{\mathrm{dR}}(U_\infty) = 0.$$

The overlap $U_0 \cap U_\infty = \mathbb{C}^\times \simeq S^1$ has

$$H^0_{\mathrm{dR}}(\mathbb{C}^\times) = \mathbb{R}, \qquad H^1_{\mathrm{dR}}(\mathbb{C}^\times) = \mathbb{R} \cdot [d\theta], \qquad H^{\geq 2} = 0.$$

The sequence with numbers. Writing δ^k for the connecting maps,

$$0 \longrightarrow H^0(\mathbb{P}^1) \xrightarrow{r_*} \mathbb{R}^2 \xrightarrow{s^0} \mathbb{R} \xrightarrow{\delta^0} H^1(\mathbb{P}^1) \longrightarrow 0 \longrightarrow \mathbb{R} \xrightarrow{\delta^1} H^2(\mathbb{P}^1) \longrightarrow 0.$$

Degree 0. $s^0(a,b) = a - b$ (constants differ on the connected overlap). Then $\ker s^0 = \{a = b\} \cong \mathbb{R}$, so $H^0(\mathbb{P}^1) = \operatorname{im} r_* = \ker s^0 = \mathbb{R}$: the sphere is connected. And s^0 is *surjective*, so $\operatorname{im} s^0 = \mathbb{R} = \ker \delta^0$, forcing $\delta^0 = 0$.

Degree 1. Exactness at $H^1(\mathbb{P}^1)$ reads $\operatorname{im} \delta^0 = \ker r_*$. Since $\delta^0 = 0$ and $r_* \colon H^1(\mathbb{P}^1) \to H^1(U_0) \oplus H^1(U_\infty) = 0$, we get

$$H^1(\mathbb{P}^1) = \operatorname{im} \delta^0 = 0.$$

Degree 2. The tail is $0 \xrightarrow{s^1} \mathbb{R} \xrightarrow{\delta^1} H^2(\mathbb{P}^1) \to 0$. The incoming s^1 has source $H^1(U_0) \oplus H^1(U_\infty) = 0$, so $\ker \delta^1 = \operatorname{im} s^1 = 0$ (δ^1 injective); the outgoing map lands in $H^2(U_0) \oplus H^2(U_\infty) = 0$, so $\operatorname{im} \delta^1 = H^2(\mathbb{P}^1)$ (δ^1 surjective). Therefore

$$\boxed{\delta^1 \colon H^1_{\mathrm{dR}}(\mathbb{C}^\times) \xrightarrow{\cong} H^2_{\mathrm{dR}}(\mathbb{P}^1)}, \qquad H^2_{\mathrm{dR}}(\mathbb{P}^1) = \mathbb{R}.$$

Diagnosis. *The entire second cohomology of $\mathbb{P}^1 = S^2$ is* created by the connecting map *out of the single winding class $[d\theta]$ on the overlap circle. The fundamental class of the sphere is the obstruction to splitting $d\theta$ across the two disks. We make this completely explicit next.*

We have recovered $H^\bullet_{\mathrm{dR}}(\mathbb{P}^1) = (\mathbb{R}, 0, \mathbb{R})$, with Euler characteristic $1 - 0 + 1 = 2 = \chi(S^2)$, consistent with $\chi(U_0) + \chi(U_\infty) - \chi(\mathbb{C}^\times) = 1 + 1 - 0$. The whole computation reads off the staircase in Figure 3.

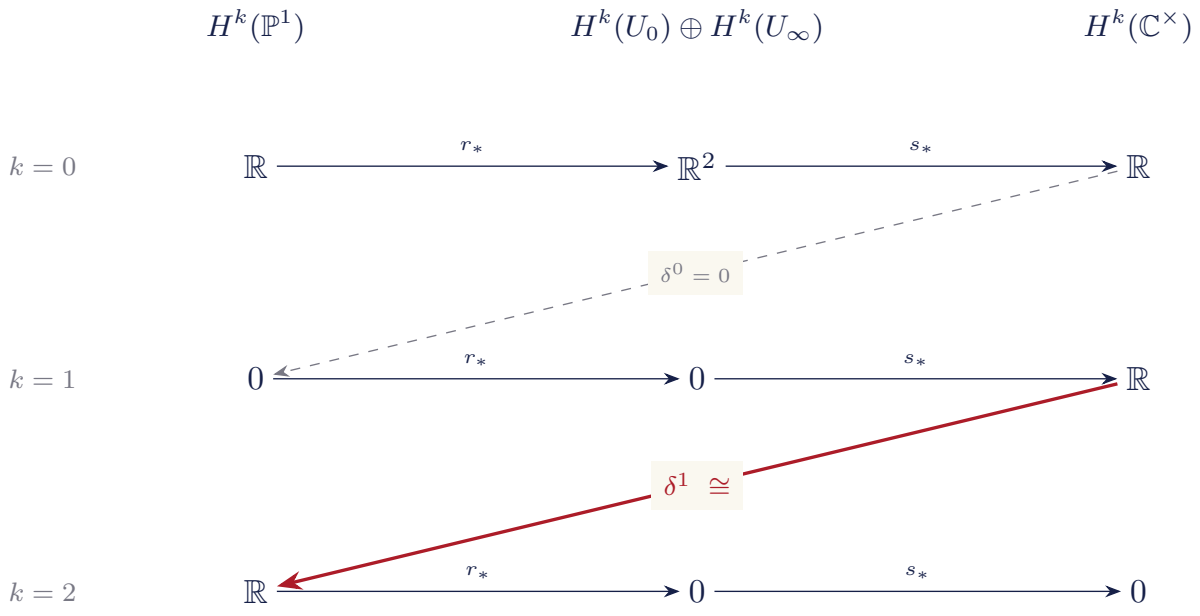


Figure 3. *The Mayer–Vietoris long exact sequence as a staircase for $\mathbb{P}^1$. Horizontal maps are restriction r_* and difference s_*; the red diagonal δ^1 is the connecting isomorphism $H^1(\mathbb{C}^\times) = \mathbb{R} \xrightarrow{\cong} H^2(\mathbb{P}^1) = \mathbb{R}$ that* manufactures *the top cohomology of the sphere out of the overlap circle, while $\delta^0 = 0$. (Supports §6, §8.)*

9 A Diagram Chase: Where the Obstruction Lives

We chase δ^1 on the generator $\gamma = \frac{1}{2\pi}\, d\theta \in \Omega^1(\mathbb{C}^\times)$ (equivalently the class of $\frac{1}{2\pi i}\frac{dz}{z}$, since $\frac{dz}{z} = \frac{dr}{r} + i\, d\theta$ and $\frac{dr}{r} = d(\log r)$ is exact). Fix a radial partition of unity $\rho_0 + \rho_\infty = 1$ with $\rho_\infty = 0$ near $z = 0$ and $\rho_\infty = 1$ near $z = \infty$. The chase

$$\gamma \in C^1 \;\rightsquigarrow\; b = (\rho_\infty \gamma,\, -\rho_0 \gamma) \in B^1 \;\rightsquigarrow\; d_B b \in B^2 \;\rightsquigarrow\; a \in A^2 = \Omega^2(\mathbb{P}^1) \;\rightsquigarrow\; [a] \in H^2(\mathbb{P}^1)$$

runs as follows.

- $\gamma \rightsquigarrow b$: $s(b) = \rho_\infty \gamma - (-\rho_0 \gamma) = (\rho_0 + \rho_\infty)\gamma = \gamma$, so b is a lift of γ (step 2 of Section 5).
- $b \rightsquigarrow d_B b$: $d_B b = (d(\rho_\infty \gamma),\, -d(\rho_0 \gamma))$. On the overlap, $d(\rho_\infty \gamma) - (-d(\rho_0 \gamma)) = d\gamma = 0$, so both components agree there: $d_B b \in \ker s = \operatorname{im} r$.

- $d_B b \rightsquigarrow a$*:* the unique global form with $r(a) = d_B b$ is

$$a = d(\rho_\infty \gamma) = d\rho_\infty \wedge \gamma \quad (\text{since } \rho_\infty d\gamma = 0),$$

  extended by 0 outside the overlap (it is supported where $d\rho_\infty \neq 0$, an annulus inside $\mathbb{C}^\times$).
- $a \rightsquigarrow [a]$*:* a is closed ($da = d(d(\rho_\infty\gamma)) = 0$), hence defines $\delta^1[\gamma] = [a] \in H^2(\mathbb{P}^1)$.

It is the generator. In polar coordinates $\gamma = \frac{1}{2\pi} d\theta$ and $\rho_\infty = \rho_\infty(r)$, so

$$\int_{\mathbb{P}^1} a = \int_{\mathbb{P}^1} d\rho_\infty \wedge \tfrac{1}{2\pi} d\theta = \int_0^\infty \int_0^{2\pi} \rho_\infty'(r)\, \tfrac{1}{2\pi}\, d\theta\, dr = \int_0^\infty \rho_\infty'(r)\, dr = \rho_\infty(\infty) - \rho_\infty(0) = 1.$$

Thus $\delta^1[\gamma]$ is the positively oriented generator of $H^2_{\mathrm{dR}}(\mathbb{P}^1)$. Figure 4 shows this construction geometrically.

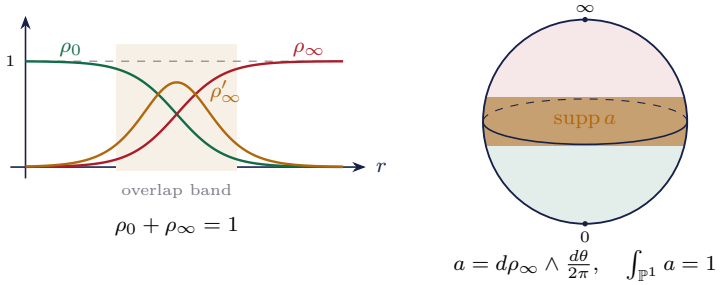


Figure 4. *The geometry of the connecting map.* Left*: a radial partition of unity $\rho_0 + \rho_\infty = 1$; the derivative ρ_∞' is a bump supported in the overlap band.* Right*: the resulting obstruction 2-form $a = d\rho_\infty \wedge \frac{d\theta}{2\pi}$ lives* only *in the equatorial band and integrates to* 1*, generating $H^2_{\mathrm{dR}}(\mathbb{P}^1)$. (Supports §9.)*

A class on the overlap $U \cap V$ may fail to come from compatible classes on U and V. The connecting map records that failure as a global cohomology class on X.

10 Relation to Gluing

Exactness is the algebra of gluing.

The dictionary:

- **kernel** = *compatibility* (data that agree where they overlap);
- **image** = *already glued* (data that genuinely come from a global object);
- **cokernel** = *obstruction* (compatible-looking data with no global origin);
- **connecting map** = the *boundary of a failed gluing attempt* (Section 9: $\delta[\gamma] = [d(\rho_\infty\gamma)]$ is literally the derivative of a partial, non-global, gluing).

Line bundles as a degree-1 Čech class. A holomorphic line bundle is glued from trivial bundles $U_\alpha \times \mathbb{C}$ by transition functions $g_{\alpha\beta} \in \mathcal{O}^\times(U_\alpha \cap U_\beta)$ satisfying the cocycle condition on triple overlaps,

$$g_{\alpha\beta}\, g_{\beta\gamma}\, g_{\gamma\alpha} = 1,$$

which is precisely the degree-1 Čech cocycle equation $\delta((g_{\alpha\beta})) = 1$ for the multiplicative sheaf $\mathcal{O}^\times$. Changing trivializations $g_{\alpha\beta} \mapsto g_{\alpha\beta} \cdot (h_\beta / h_\alpha)$ with $h_\alpha \in \mathcal{O}^\times(U_\alpha)$ alters the cocycle by a *coboundary*. Hence isomorphism classes of line bundles form

$$\boxed{\mathrm{Pic}(X) = \check{H}^1(X, \mathcal{O}^\times)}.$$

Computing $\mathrm{Pic}(\mathbb{P}^1)$ by a long exact sequence. The exponential sheaf sequence on the complex manifold $\mathbb{P}^1$,

$$0 \longrightarrow \mathbb{Z} \xrightarrow{2\pi i} \mathcal{O} \xrightarrow{\exp} \mathcal{O}^\times \longrightarrow 0,$$

is exact: exp is surjective *as a sheaf map* (locally every nonzero holomorphic function has a logarithm on a small disk, though *not* globally), and its kernel is the locally constant sheaf $2\pi i\,\mathbb{Z}$. Its long exact sequence reads

$$\cdots \longrightarrow H^1(\mathbb{P}^1,\mathbb{Z}) \longrightarrow H^1(\mathbb{P}^1,\mathcal{O}) \longrightarrow H^1(\mathbb{P}^1,\mathcal{O}^\times) \xrightarrow{c_1} H^2(\mathbb{P}^1,\mathbb{Z}) \longrightarrow H^2(\mathbb{P}^1,\mathcal{O}) \longrightarrow \cdots$$

Now plug in the values computed in Section 11 (Examples C–D) and standard topology: $H^1(\mathbb{P}^1,\mathbb{Z}) = 0$ and $H^1(\mathbb{P}^1,\mathcal{O}) = 0$, while $H^2(\mathbb{P}^1,\mathbb{Z}) = \mathbb{Z}$ and $H^2(\mathbb{P}^1,\mathcal{O}) = 0$. The sequence collapses to

$$0 \longrightarrow \mathrm{Pic}(\mathbb{P}^1) = H^1(\mathbb{P}^1,\mathcal{O}^\times) \xrightarrow{c_1} H^2(\mathbb{P}^1,\mathbb{Z}) = \mathbb{Z} \longrightarrow 0,$$

so c_1 is an isomorphism:

$$\boxed{\mathrm{Pic}(\mathbb{P}^1) \cong \mathbb{Z}, \qquad \mathcal{O}(d) \longleftrightarrow d = c_1}.$$

Diagnosis. *The connecting map of the exponential sequence* is *the first Chern class / degree. The same integer that appeared as the transition exponent $g_{0\infty} = z^{-d}$ in the divisor ledgers is here produced cohomologically as the obstruction to taking a global logarithm of a transition function.*

Figure 5 pictures why that global logarithm fails.

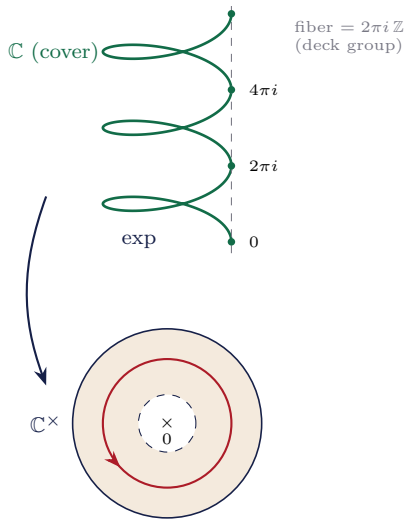


Figure 5. *The exponential sequence $0 \to \mathbb{Z} \to \mathcal{O} \xrightarrow{\exp} \mathcal{O}^\times \to 0$ made visible. The helix is the cover $\mathbb{C}$;* exp *winds it down onto $\mathbb{C}^\times$ with fiber $2\pi i\,\mathbb{Z}$ (the constant sheaf $\mathbb{Z}$). A loop in $\mathbb{C}^\times$ lifts to a path that fails to close by a multiple of $2\pi i$, so the obstruction to a global logarithm is exactly the winding number — read by the connecting map as $c_1 = \deg$. (Supports §10.)*

11 Worked Micro-Examples

Example A: S^1 — a disconnected overlap creates H^1

Cover S^1 by two arcs U, V, each contractible, with $U \cap V$ *two* disjoint arcs $\simeq \{p\} \sqcup \{q\}$. Then $H^0(U) \oplus H^0(V) = \mathbb{R}^2$ and $H^0(U \cap V) = \mathbb{R}^2$ (one $\mathbb{R}$ per component). Mayer–Vietoris:

$$0 \to H^0(S^1) \to \mathbb{R}^2 \xrightarrow{s} \mathbb{R}^2 \to H^1(S^1) \to \underbrace{H^1(U) \oplus H^1(V)}_{=0}.$$

With constants (a, b) on (U, V), the difference on each overlap component is the same value $a - b$, so

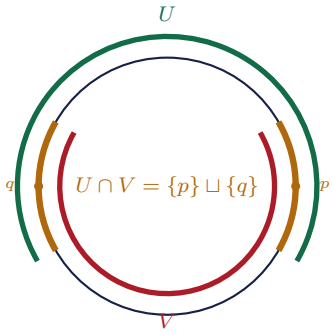
$$s(a,b) = (a-b,\ a-b), \qquad \ker s = \{a = b\} \cong \mathbb{R}, \qquad \mathrm{im}\, s = \{(t,t)\} \cong \mathbb{R}.$$

Hence $H^0(S^1) = \ker s = \mathbb{R}$ (connected), and

$$H^1(S^1) \cong \operatorname{coker} s = \mathbb{R}^2/\{(t,t)\} \cong \mathbb{R}.$$

Diagnosis. *The H^1 of the circle is exactly the "extra" $\mathbb{R}$ in $H^0(U \cap V) = \mathbb{R}^2$ caused by the overlap being* disconnected. *The second overlap component is where a global* 1*-cocycle hides.*

Figure 6 makes the disconnected overlap visible.



two overlap components $\Rightarrow H^0(U \cap V) = \mathbb{R}^2 \Rightarrow H^1(S^1) = \mathbb{R}$

Figure 6. *Mayer–Vietoris for S^1 with two arcs U (outer, green) and V (inner, red). Their intersection has* two *components p, q (amber); the "extra" copy of $\mathbb{R}$ in $H^0(U \cap V) = \mathbb{R}^2$ is exactly what the connecting map turns into $H^1(S^1) = \mathbb{R}$. (Supports Example A.)*

Example B: S^2 — the overlap circle detects the fundamental class

Cover S^2 by two disks U, V with $U \cap V \simeq S^1$ (an equatorial band). This is the same shape as the $\mathbb{P}^1$ cover. The relevant segment is

$$\underbrace{H^1(U) \oplus H^1(V)}_{0} \to H^1(S^1) = \mathbb{R} \xrightarrow{\delta} H^2(S^2) \to \underbrace{H^2(U) \oplus H^2(V)}_{0},$$

so $\delta\colon H^1(S^1) \xrightarrow{\cong} H^2(S^2)$, giving $H^2(S^2) = \mathbb{R}$ and

$$\boxed{H^2(S^2) \cong H^1(S^1)}.$$

Diagnosis. *The generator of $H^2(S^2)$ is the connecting image of the generator of H^1 of the equatorial circle: the sphere's area class is the obstruction to extending the circle's winding form over both caps (Section 9 makes the form $d\rho_\infty \wedge \frac{d\theta}{2\pi}$ explicit).*

Figure 7 summarizes this single step.

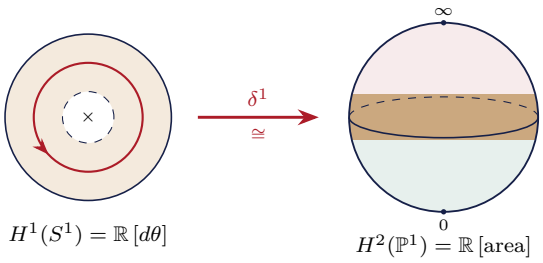


Figure 7. *The whole document in one picture: the connecting isomorphism δ^1 carries the winding class on the overlap circle (left) to the area class supported in the equatorial band of the sphere (right). The obstruction to extending the circle's* 1*-form over both caps* is *the fundamental class of $\mathbb{P}^1 = S^2$. (Supports Example B; summarizes §7–§9.)*

Example C: $\mathbb{P}^1$ holomorphically — dz/z and winding

Use $U_0 = \mathbb{P}^1 \setminus \{\infty\}$, $U_\infty = \mathbb{P}^1 \setminus \{0\}$, $U_0 \cap U_\infty = \mathbb{C}^\times$. The overlap carries the meromorphic form dz/z, whose period

$$\frac{1}{2\pi i}\oint_{|z|=1} \frac{dz}{z} = 1$$

records *winding around* 0. Two computations live on this overlap:

- *Constant sheaf / topology:* $[\,\frac{1}{2\pi i}\frac{dz}{z}\,]$ generates $H^1_{\mathrm{dR}}(\mathbb{C}^\times)$ and, via δ, generates $H^2(\mathbb{P}^1)$ (Sections 8–9).
- *Coherent sheaf $\mathcal{O}$:* the Čech computation $\check{H}^1(\{U_0, U_\infty\}, \mathcal{O})$ uses Laurent series. Every Laurent series $\sum_{n\in\mathbb{Z}} c_n z^n$ on $\mathbb{C}^\times$ splits as

  $$\sum_{n\le 0} c_n z^n \ \in \mathcal{O}(U_\infty) \quad \text{minus} \quad \Big(-\sum_{n\ge 1} c_n z^n\Big) \in \mathcal{O}(U_0),$$

  so δ is surjective and $H^1(\mathbb{P}^1, \mathcal{O}) = \operatorname{coker}\delta = 0$. Likewise $\ker\delta = \{\text{global holomorphic}\} = \mathbb{C}$ (Liouville), so $H^0(\mathbb{P}^1, \mathcal{O}) = \mathbb{C}$.

Diagnosis. *The* same *overlap $\mathbb{C}^\times$ feeds two different sheaves: against $\mathbb{R}$ (topology) it produces the nonzero class detecting $H^2(\mathbb{P}^1)$; against $\mathcal{O}$ (analysis) the Laurent splitting kills H^1. The discriminating data is the coefficient sheaf, not the space.*

Example D: Line bundles — cocycle, coboundary, and $\check{H}^1(\mathcal{O}^\times)$

A line bundle on $\mathbb{P}^1$ is given by a single transition function $g_{0\infty} \in \mathcal{O}^\times(\mathbb{C}^\times)$ (two charts, so the cocycle condition $g_{\alpha\beta}g_{\beta\gamma}g_{\gamma\alpha} = 1$ reduces to $g_{\infty 0} = g_{0\infty}^{-1}$; triple overlaps are empty). Now

$$\mathcal{O}^\times(\mathbb{C}^\times) = \{\, z^n e^{h(z)} : n \in \mathbb{Z},\ h \in \mathcal{O}(\mathbb{C}^\times)\,\},$$

and the coboundaries are f_∞/f_0 with $f_0 \in \mathcal{O}^\times(U_0) = \{e^{h_0}\}$, $f_\infty \in \mathcal{O}^\times(U_\infty) = \{e^{h_\infty}\}$ (a nowhere-zero holomorphic function on a contractible chart is an exponential). Splitting the Laurent series $h = h_0 + h_\infty$ as in Example C, e^h is a coboundary, so

$$g_{0\infty} = z^n e^h \ \sim\ z^n,$$

and the class is the integer

$$n = \frac{1}{2\pi i}\oint_{|z|=1} \frac{dg_{0\infty}}{g_{0\infty}} = \frac{1}{2\pi i}\oint_{|z|=1} d\log g_{0\infty}.$$

Hence $\check{H}^1(\mathbb{P}^1, \mathcal{O}^\times) \cong \mathbb{Z}$ (matching Section 10's exponential LES). With the convention $g_{0\infty} = z^{-d}$ of the divisor ledgers, $n = -d$ and

$$\deg \mathcal{O}(d) = -n = -\frac{1}{2\pi i}\oint_{|z|=1} d\log g_{0\infty} = d.$$

Diagnosis. *The cocycle condition is the closedness ($\delta = 1$) of a degree-1 multiplicative cochain; the coboundary freedom is exactly the change-of-trivialization h_β/h_α. The only surviving invariant is the winding number of the transition function — the degree.*

Figure 8 shows the cocycle and the $\mathbb{P}^1$ gluing.

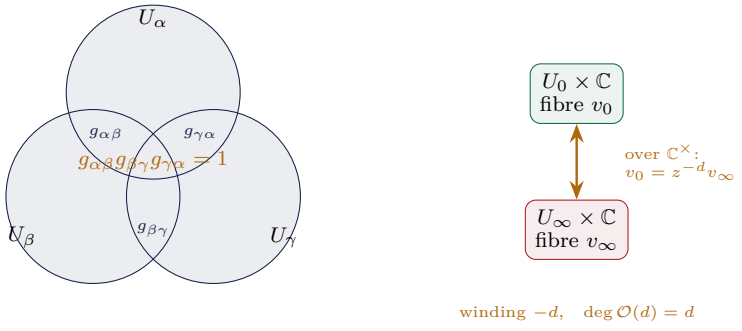


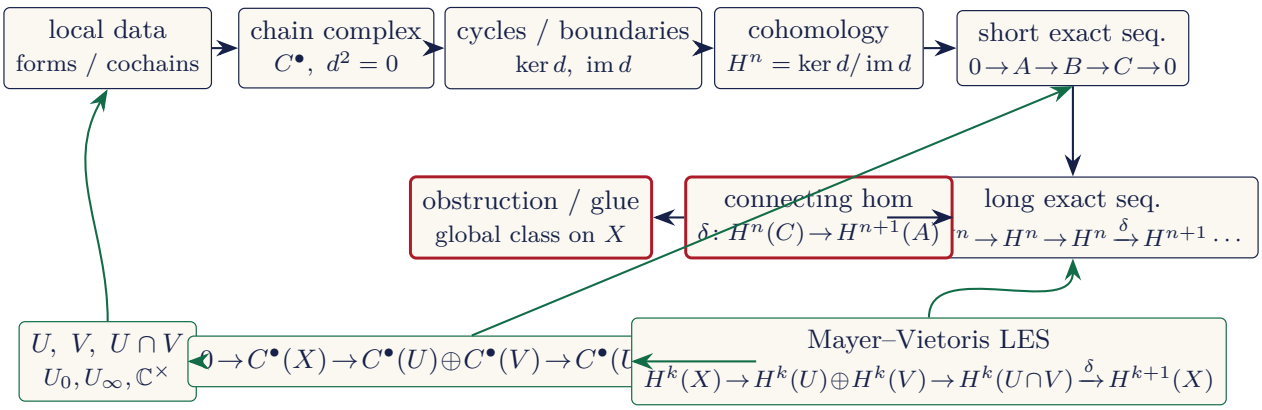
Figure 8. *Line bundles as a degree-1 Čech class.* Left*: on a triple overlap the transition functions obey the cocycle condition* $g_{\alpha\beta}g_{\beta\gamma}g_{\gamma\alpha} = 1$. Right*: on* $\mathbb{P}^1$ *the two trivial pieces are glued over* $\mathbb{C}^\times$ *by* $g_{0\infty} = z^{-d}$*, whose winding number* $-d$ *is the only invariant, giving* $\deg \mathcal{O}(d) = d$*. (Supports §10, Example D.)*

12 Summary Table

Tool	Input data	Algebraic form	What it measures	Concrete meaning here
chain complex	forms / cochains & a differential d	$C^\bullet,\ d^2 = 0$	the apparatus in which "closed" and "exact" make sense	$\Omega^\bullet(\mathbb{P}^1)$; Čech $\check{C}^\bullet(\{U_0, U_\infty\}, \mathcal{F})$
cohomology	a chain complex	$H^n = \ker d / \operatorname{im} d$	closed modulo exact (genuine obstructions)	$H^\bullet_{\mathrm{dR}}(\mathbb{P}^1) = (\mathbb{R}, 0, \mathbb{R})$; $H^\bullet(\mathbb{P}^1, \mathcal{O}) = (\mathbb{C}, 0)$
short exact sequence	sub/quotient of complexes	$0 \to A \to B \to C \to 0$	comparison of three worlds (global, local sum, overlap)	$0 \to \Omega^\bullet(X) \to \Omega^\bullet(U_0) \oplus \Omega^\bullet(U_\infty) \to \Omega^\bullet(\mathbb{C}^\times) \to 0$
connecting hom	a SES of complexes	$\delta\colon H^n(C) \to H^{n+1}(A)$	obstruction to lifting a closed class from C to B	$\delta[d\theta]$ = generator of $H^2(\mathbb{P}^1)$
long exact sequence	a SES of complexes	$\cdots \to H^n \to H^n \to H^n \xrightarrow{\delta} H^{n+1} \to \cdots$	how the three cohomologies constrain each other	computes $H^\bullet(\mathbb{P}^1)$; computes Pic
Mayer–Vietoris	a cover $X = U \cup V$	the LES of the cover SES	global cohomology from local pieces + overlap	two disks $\cup$ annulus rebuild $S^2 = \mathbb{P}^1$
Čech cocycle	cover & sheaf $\mathcal{F}$	$\{g_{\alpha\beta}\},\ \delta g = 0$	locally trivial data compatible on overlaps	transition $g_{0\infty} \in \mathcal{O}^\times(\mathbb{C}^\times)$ of a line bundle
coboundary	a 0-cochain	$g_{\alpha\beta} = h_\beta / h_\alpha$	data that is a re-trivialization, hence trivial	change of chart e^{h_∞}/e^{h_0}; winding 0
gluing obstruction	cocycle mod coboundary	$H^1(X, \mathcal{F})$	failure of local data to be global	$\check{H}^1(\mathcal{O}^\times) = \mathrm{Pic}(\mathbb{P}^1) = \mathbb{Z}$ (the degree)

13 Final Conceptual Diagram

The whole pipeline, with the Mayer–Vietoris branch feeding the short exact sequence that drives everything.



Reading. The top row is the abstract machine: local data assemble into a chain complex; the differential carves out cycles and boundaries; their quotient is cohomology; a comparison of complexes

is a short exact sequence; its shadow on cohomology is the long exact sequence, whose connecting map is the obstruction that becomes a global class. The green Mayer–Vietoris branch is the geometric instantiation: cutting $\mathbb{P}^1$ into U_0, U_∞ with overlap $\mathbb{C}^\times$ is *exactly* the short exact sequence that drives the machine, and its connecting map manufactures $H^2(\mathbb{P}^1)$ and (via $\mathcal{O}^\times$) the degree of a line bundle.

Homological algebra is not formal machinery. It is a controlled way to watch local data become global structure — and to measure, as a single cohomology class, precisely where the gluing fails.